

Recommersion: Context-Aware Vocal Assistant for Emotion-Driven Music Recommendations Using Real-Time User Feedback

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Abstract—*Recommersion* is a context-aware music recommender system conceived with a vocal assistant and an adaptive real-time emotion recognition support. Speech Emotion Recognition (SER) models extract valence and arousal dimensions by analyzing users vocal inputs, flowing into a similarity criterion for creating an intended playlist by selecting songs in music emotion-driven datasets such as DEAM, PMemo and SpotiGeM. The management of the aforementioned features are allowed by a real-time graphical interface. 19 participants were involved for evaluating playlist relevance, each generated by two SER models, *Audeering Wav2vec 2.0* based model and a custom-built alternative, joined together with Euclidean and Cosine similarity conditions. Results demonstrated that *Audeering* and Euclidean combination provided on average the most accurate recommendations, achieving a Mean Average Precision (MAP) of 0.787, a Mean Reciprocal Rank (MRR) of 0.816 and a Normalized Discounted Cumulative Gain (NDCG) of 0.842 with SpotiGEM data integration. Both models showed minimal gender discrepancies in emotion predictions, despite a lower testing performance regarding the Mel feature-based customized model. Future scenarios are promising, especially in applications including automotive environments and personalized entertainment. On the other hand, aiming to enhance contextual precision through text sentiment analysis and increment data cardinality with new recommendation and ranking strategies.

I. STATE OF THE ART

A. Emotion Perception and Context Recognition

Nowadays, emotion and context recognition interplay creates personalized and meaningful interactions between users and everyday technology: they profoundly influence decision-making, preferences and behavior, making their accuracy important for systems designed to respond empathetically and effectively like vocal assistants. For this work, this specific combination ensures that the vocal assistant tries to offer relevant recommendations but particularly emotionally resonant, creating personalized experiences in daily specific moments. Based on Barret [1, 2], visual scenes, voices, faces, cultural orientation and even words shape how emotion is in general perceived. Its suggestion tells that an human emotion it is not just recognizable from a face: for that matter, Carrol [3] pointed out that most of the times a portrait is not enough to judge the emotion correctly. Moreover, the stimulus is deemed

into a set of coordinated behavioral responses, allowing individuals to comprehend which kind of mood the examined person is feeling. For example, language role is fundamental in terms of perception [2]: in fact, emotion words can influence how faces are perceived. When faces, which could be a blend of happiness and anger, are shown alongside the word "angry," they tend to be interpreted as expressing more anger. This effect becomes even more pronounced when participants are asked to explain why some portraits appear angry [4]. Language is strongly affected by the culture the individual is belonged to [5]: Balkwill and Thompson [6] proposed a communication model in which the emotions are correlated by a mixture of cultural and universal marks, bringing the listener to try to identify them in a completely different tonal system. Specifically, thirty western auditors assessed the principal emotions (sadness, anger, joy and peace) in "12 Hindustani raga" system. Furnishing also psychological cues like rhythmic and melody complexity, listeners were able to distinguish their own emotional sensibilities on the music proposed. A specific case in music instead is explored by one study of University of Toronto [7] on how listeners perceive emotions in this environment compared to their emotional reactions to the same pieces. Participants rated happiness and sadness both as expressed by the music and as personally felt, using identical scales. The music varied in tempo (fast or slow) and mode (major or minor). While perception and feeling ratings were closely related, perception ratings were generally higher, especially for music with clear emotional cues (e.g., fast-major for happiness or slow-minor for sadness) and for sad music overall. Listeners' feelings were influenced by how they perceived the emotions conveyed by the music. Fast tempos and major keys were associated with higher happiness ratings, slow tempos and minor keys with higher sadness ratings, and mixed emotional cues (fast-minor or slow-major) led to combined feelings of happiness and sadness. Additionally, listeners showed a sense of ambivalence toward sad-sounding music. Music expressions often transmit emotions through performance: this paper of Gabrielsson, A., & Lindström, E. (2001) [8] clarify that the perception is a sum of the emotions inherent in the music composition and those of the actual ex-

ecution. Moreover, in another study guitarists were successful when utilizing acoustic cues for emotion communication [9]. A coherent survey examined listeners' cue utilization based on two experiments, explaining how they relate with them for decoding articulations [10]. In the first experiment, listeners assessed the emotional expression conveyed in performances, and findings revealed several key insights. First, participants were able to accurately identify the intended emotional expressions from synthesized performances, with a comparable level of accuracy to live performances. However, reversing the sequence of performances impacted their accuracy more significantly, highlighting their stronger reliance on prosodic contours. In a subsequent testing, five factors (tempo, sound level, spectrum, articulation, and attack) were systematically manipulated using a factorial design. Participants rated the emotional expression of different combinations of these aspects using adjective scales, and multiple regression analysis was employed to model their judgment patterns. The analysis revealed that listeners utilized all available cues in their evaluations, though none of the cues provided absolute certainty in their influence. Moreover, the relative importance of each cue varied significantly depending on the emotion being assessed.

In general, several studies have shown that transmit musical performance emotion belong to the importance of acoustic cues. Friberg et. al [11] developed an automatic prediction system, starting from audio feature extraction to exploit linear regression models, which denote accurate predictions on the intended emotion of the player. Generally, features in the music environment are conceived as low-level ones, like spectral (MFCCs, Zero Crossing Rate) or psycho-acoustic (beat strength and rhythm). Theory dissects music and provide us certain schemes to analyze upon it, but in the majority of cases a random listener should not be able to detect certain attributes. This points us that humans focus more on high level features, including genre or the feeling the song could offer. Friberg et al. [12] proposed in 2014 a novel approach to Music Information Retrieval (MIR) by introducing listener-based perceptual features as intermediate representations. Traditional MIR methods rely heavily on low- and mid-level audio features like spectral properties and music theory constructs (e.g., pitch, rhythm). However, these often fail to capture the nuanced ways humans perceive music. The authors selected nine perceptual features inspired by psychological and ecological music studies, including speed, rhythmic clarity and harmonic complexity. These features were evaluated through two listening experiments involving music clips and polyphonic ringtones. Results demonstrated that emotional responses, particularly dimensions like energy and valence, could be reliably predicted using perceptual features, with explained variances reaching up to 93%. It also means that perceptually-based models conduct to a depth comprehension of human vision.

B. Emotion dealing in Music Platforms

Plenty of music platforms offer the opportunity to group thousands of songs into emotional moods (either for focusing

or just relaxing) or musical genres automatically. Context-aware music recommender systems (CARS) have gained significant interest: due to their ability, they could enhance music recommendations by integrating information such as time of day, user location or emotional state. This efficient literature review [13] composed by Lozano Murciego *et al.* analyzes existing research on CARS in the music domain. Their focus highlights key contextual factors, data collection devices, integration methodologies, evaluation protocols, datasets and future research directions. Music recommendation systems [14] present unique challenges that distinguish them from other domains like travel, literature or cinema. A trivial difference lies in the time required to experience the recommended content: for example, reading a book can take days or even weeks. Watching a film requires hours. Instead, listening to a song typically lasts only a few minutes: this brevity allows users to form opinions quickly, making them more likely to skip tracks they dislike. Another distinction is its repetitive nature: unlike films or books, which are rarely revisited shortly after being experienced, songs can be played repeatedly (even within the same day) without diminishing user satisfaction. This gives shape to a unique dynamic where recommending previously played songs can be perceived in a positive way. The consequences of poor recommendations in music are relatively minor: users can easily skip unsatisfactory tracks with just a click, making them more forgiving of less accurate suggestions. Furthermore, these same recommendations can target various levels of specificity, starting from individual tracks and playlists to entire albums or artists, adding great flexibility. For regular music listeners, content-based recommendation approaches can be highly effective. However, research in music psychology [15] suggests that short-term musical preferences are often shaped by various contextual factors, including emotional states, physical surroundings and social environments. Specific types of music are commonly associated with particular moods, activities or even specific times of day and locations. Thus, integrating contextual information into music recommendation systems has emerged as a crucial challenge. To conclude, this paper [14] highlights the exploration of better emotion-aware recommender systems in one of the relevances for future research. It connotes a significant impingement on users' musical tastes, exploiting voice assistants in particular. An affective vocal burst (A-VB) competition [16] was conducted in order to comprehend emotional behaviors via non-verbal human sounds. This challenge was split into 4 individual tasks:

- A-VB-HIGH: regression task with CCC metric for predicting levels of 10 high-level emotions like fear or surprise;
- A-VB-TWO: same task as the point above for valence and arousal;
- A-VB-CULTURE: multi-output regression task in which 40 combinations of cultural provenience and emotions are analyzed using CCC;
- A-VB-TYPE: multi-classification task for vocal expres-

sion (laugh, scream, cry etc.) utilizing Unweighted Average Recall (UAR) on 8 classes.

Chonnam National University took part in this competition: they developed a fine-tuning on 4 different versions of Wav2vec network [17], binding it with basic neural networks and a pooling method. The researchers of this university observed the simplicity of this architecture, outperforming the baseline of the organizer on the test set.

Deezer streaming service generates mood-inspiring individual playlists, which its algorithm *Flow Moods* benefits from collaborative filtering, audio content analysis and mood annotations from professional music curators [18]. A work of Lu and Liu [19] detects mood automatically by extracting features such as intensity, timbre and rhythm, with the avail of psychological aspects of western habit. Tzanetakis developed a music genre classification inspecting audio signals, examining the acoustic attributes mentioned before.

One of the MIR state-of-the-art approach is *MusicSense* [20], a contextual model recommendation system which processes Web documents that the user is reading. Then, it compares the emotion expressed in the document corpus with the songs examined. These key data distribute over the emotion mixture by means of the generative part, called *Emotional Allocation Modeling*.

InCarMusic [21] recommends songs based on traffic conditions or the driver's mood. Baltrunas et al. designed a system in which users were asked to judge if a context situation around them is determinant for their decision making and rating, proving in closing that is conceivable to construct a valid context-aware mobile recommender system. Moreover, Ayça Berfu Ünal pointed out the music's impact on driving performance [22]. According to her research, the psychologist in environment and traffic discovered that music helps the car conductors in their ride. For this reason, the drivers are more concentrated: the music stimulation triggers the brain functions regarding alertness or to preventing boredom. In addition, opposed to the belief of the majority of people, a challenge task like reverse parking is not impaired by music. Instead, the concentration is more supported. In conclusion, she observed that the background listening can be more distracting, leading to necessitate more mental effort.

C. Valence and Arousal Real-Time Manipulation

Emotion has proven to be a valuable and effective tool for classifying music. The concept behind sense representation assumes that emotions can be viewed as continuous variables, which can be mapped onto a range of real numbers. One of the early attempts to systematically describe human emotions was made by Russell [23]: he introduced a circumplex model, in which emotions are represented across two bipolar dimensions: valence, (or pleasantness, ranging from pleasant to unpleasant), and arousal (ranging from high arousal to low arousal, or sleep). Each emotion can thus be characterized as a combination of these two aspects. Building on Russell's framework, Thayer [24] adapted the model specifically for music, redefining the two core dimensions as "arousal" and

"valence." In Thayer's adaptation, the arousal dimension spans from calm to energetic, while valence dimension ranges from negative to positive. This allowed emotion descriptors in music to be mapped onto a spectrum from silent to energetic (arousal) and from negative to positive (valence). A Real-Time Music Emotion Modulation System for Soundtracks was developed by The Hong Kong University of Science and Technology [25]. The purpose of this system is to examine and modify levels of valence and arousal in real-time, designing an automatic production of soundtracks based on what the user is experiencing. The modulation of the dimensional values is correlated to the music parameters (Tempo, Dynamics, Pitch and Mode). The results have shown that "The Emotion Modulator" is conditioned by the inspected musical genre, leading to work well on classical and thematic pieces and less in modern music. Plenty of studies focus principally on primary musical cues [8, 9, 26, 27, 28]: specifically, the study by Eerola, Friberg, and Bresin [27] attempts to understand how these parameters influence emotional expression in music. The research systematically manipulated six primary musical cues (mode, tempo, dynamics, articulation, timbre and register) across various levels and analyzed their contributions to perceived emotions (happy, sad, peaceful, scary). Mode was the most influential cue, followed by tempo, register, and dynamics, with their importance varying across emotions. Most cues showed linear relationships with emotional ratings, explaining 77–89% of the variance, though some modest quadratic effects were observed for timbre and dynamics. The cues predominantly operated in an additive manner, meaning their effects combined independently without significant interactions. These findings support the idea that musical cues mimic emotional expressions in speech, aligning with physiological state theories. The study highlights the structured and predictable influence of musical cues on emotional expression, using a factorial design to isolate their contributions. Valence and arousal are proper indicators for analyzing and describing the gait of the emotion during time. These features are fundamental when activities like gaming or virtual reality are assessed. A Real-Time Adaptation for Music Emotion [29] was investigated, where music is adjusted based on the emotion of the in-game state. The comprehension of a metric called 2 Dimensional Emotion Space (2DES) in octal form, collated with music principal rules (tempo, pitch etc.) establish a new octal set called *primary music emotion structural rules*. The particularity of 2DES is its continuous numerical form and the strong bond to the Hevner adjective circle [30] response, leading to effective translations owing to the aid of musical cues following Schubert work [31]. Thus, it gave life to an early dynamic model: first results implied that the carried out rules have an effective outcome on 3 out of 4 quadrants (happier, sadder and dreamier). Building upon these foundational studies, the integration of emotion recognition and the manipulation of valence and arousal in music presents a compelling case for further development. By leveraging this dual-axis model, systems can dynamically fit auditory experiences to align with what the users feel, creating

a more engaging interaction. This is particularly significant in areas such as personalized entertainment, therapeutic interventions and productivity enhancement, where music’s emotional resonance can profoundly influence outcomes.

For example, in gaming and virtual reality worlds is really significant: the ability to dynamically adapt music in response to users’ evolving emotions adds depth and immersion and enhances the overall experience. Beyond entertainment, such systems hold promise in mental health applications, where personalized soundscapes can aid in stress management or mood regulation, or they can use emotion-aware music adaptations to sustain focus or encourage relaxation. The potential of these applications underscores the importance of developing systems that harness the emotional power of music in a nuanced and context-sensitive manner.

D. Similar Works

The development of systems capable of recognizing and responding to emotions is an arduous challenge: in the domain of SER, the advancements in using deep neural networks have significantly enhanced emotion prediction by modeling valence and arousal in a two-dimensional space. Yiang and Hirschberg [32] elaborated a parallel CNN task for retrieving time and mel characteristics from a raw speech waveform. The data from the training process are collected by RECOLA [33] and SEMAINE [34] datasets. The combined result is then passed into a 2-layer Bidirectional LSTM to shape the contextual substance, reflecting its growing importance in emotion modeling.

Furthermore, the boost in transformers-architecture development have led to a considerable number of machine learning experiments. Wagner *et al.* [35] from *Audeering* peculiarly explored the application of *wav2vec 2.0* and *HuBERT* for dimensional Speech Emotion Recognition (SER) tasks. It emphasizes closing the “valence gap”, a persistent challenge in SER, by achieving state-of-the-art performance for valence prediction of 0.638 on the MSP-Podcast [36] dataset. The research also evaluates the impact of model size, pre-training data, fine-tuning strategies and generalization abilities on downstream performance. This work demonstrated a strong cross-corpus generalization using IEMOCAP [37] and MOSI datasets and robustness under various augmentations, comparing to the traditional CNN baseline. Fairness on gender and speaker individual was particularly highlighted, which are absolutely not trivial in a SER environment. The model has a pre-trained layer, *wav2vec 2.0* or *HuBERT* (pre-trained on diverse datasets like LibriSpeech, Libri-Light and VoxPopuli, covering a range of domains such as audiobooks, conversational speech, parliamentary recordings), tested both in base and large variants. Then, the fine-tuning process was performed on MSP-Podcast with a CCC loss function [38]: thanks to its properness for the correlation-based metric and the bias into Pearson’s correlation coefficient (CC), this regression-based metric is commonly used for dimensional SER [39, 40, 41, 42]. The calculation is:

$$\rho_c(CCC) = \frac{2\rho\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2 + (\mu_x - \mu_y)^2}$$

where:

- σ_x^2 and σ_y^2 are the variances for x and y, whereas σ_x and σ_y are the corresponding covariances;
- μ_x and μ_y are the means for the two variables.
- ρ is the correlation coefficient between x and y.

Hence, the calculation of the respective loss is:

$$CCC_{loss} = 1 - \rho_c$$

Only the transformer layers of the pre-trained model were fine-tuned, while the CNN feature extractor layers were frozen to focus on high-level feature adaptation. Finally, the model outputs valence, arousal and dominance and calculates a weighted average loss for optimizing all dimensions simultaneously. In this experiment their best model [43] has been used for research and test in the recommendation field.

Aside from advancements in SER, the adapting of context-aware systems has been studied by Byeong-jun *et al.* (2010) [44], which proposed a context-aware recommendation system based on human emotional states. The contributions of this work are the following:

- 1) Emotion State Transition Model (ESTM): this component tries to identify and recommend music that effectively transitions users to their desired emotional state. The model uses a mathematical representation, principally in the form of a vector composed of weighted emotion states at specific time points. The combination of the vectors of the initial and final emotional states constructs an emotion state transition;
- 2) COMUS: ontology represented in OWL language, which extracts high-level music metadata (e.g., musical features, genre, instrument, taxonomy, mood and situation) and correlates them with a person’s daily moments and contextual information. This ontology enables the evaluation of the user’s desired emotional state.
- 3) Recommendations were based on a classification of low-level musical features. Using a beat-tracking algorithm, the system extracted various acoustic features for each frame, including novel features such as beat lengths and chord/scale variations.

In summary, the estimation of emotional states performed by combining COMUS ontology, ESTM model and low-level musical feature mapping through SVM significantly improves recommendation accuracy.

Further extending these capabilities, a real-time interactive music software like “Emotional Equalizer” [45] was designed. From suggesting themes for film music composers to gaming background music in accordance with the player moves, this emotion-based system can modulate the mood of the music while conserving the underlying melody, demonstrating the potential of emotion-driven music generation. The interface permits to adjust through sliders 8 moods parameters, like sadness or romance, with the additional possibility to change accompaniment and instrumentation and to schedule the different moods across time. The score is generated with conversion

of the emotion mood into music parameters like tempo and mode, defining harmony and melody.

”Recommersion” synthesizes these approaches into a unified framework, bridging emotion-aware computing, context-aware recommendation systems and real-time user feedback. By integrating SER capabilities for vocal emotion recognition, context-sensitive recommendations and dynamic emotional re-calibration via valence and arousal, this framework represents a significant step forward in personalized, adaptive and holistic user experiences. The gathering of these components results in a vocal assistant that, from a contextual emotional user input, recommends tracks dynamically in order to better align with users’ evolving emotional needs.

II. SYSTEM OVERVIEW

Recommersion is a vocal assistant capable of contextualizing a specific situation based on user-provided inputs influenced by their emotional state. More deeply, it is a Context-Aware Recommender System (CARS) designed to suggest almost 20’000 songs from DEAM [46], PMemo [47] and SpotiGEM [48] datasets. These recommendations are based on emotional dimensional values (valence and arousal) captured through speech or manually adjusted via sliders with the avail of a GUI (see II-E). The system leverages Speech Emotion Recognition (SER) models, including the Model for Dimensional Speech Emotion Recognition based on Wav2vec 2.0 by *Audeering* [43] and a custom built model.

A. Architecture

The system processes the user’s input (either vocally or via sliders) through a basic Recommender System, which uses one of the Speech Emotion Recognition (SER) models and a given similarity criterion (Euclidean or Cosine) to suggest an optimal number of tracks submitted by the user. Once the recommendation is generated and a musical fragment begins playing, users can manage a proper interface for song’s reproduction real-time adjustments. They can also select any track from the generated playlist. If the recommendation does not satisfy the utilizer, it can be easily recomputed in the same way, opting for a more accurate indication over sliders. As you can deduce from figure II-A, if any user find the first songs too happy for their mood, they can decrement meticulously the valence parameter. Then, the submission of the updated parameters is recalculated and driven forward to the system.

B. Custom Model

The objective of this model is to get as close as possible to the lower bound achieved by the *Audeering* emotion dimension model [43]. The metrics accomplished by the model just mentioned on the test set are depicted in Table I:

A first drafting was designed by implementing a *Wav2vec Feature Extractor* of the input speech waveform, followed by padding to ensure uniform length of 6 seconds. Then, speech

Fig. 1. General architecture of Recommersion. The user has initially access to the microphone or sliders for computing the emotion-dimensional values. Then, through a similarity criterion, the songs are reordered and trimmed down to the number of songs originally desired. Once the reproduction has started, the utilizer can approach music player’s tools or recompute the recommendation once again.

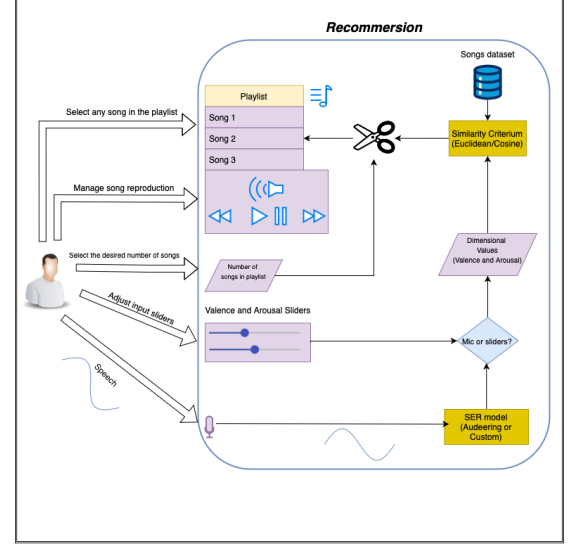


TABLE I
Audeering MODEL’S METRIC RESULTS IN THE TEST SET.

Metrics Results (<i>Audeering</i> Model)		
Metric	Valence	Arousal
CCC Loss	0.459	0.405
L1	0.130	0.118
L2	0.029	0.024
R ²	0.068	0.154

travels onto the pre-trained Wav2vec2.0-base net, which contains approximately 85M parameters. The CNN extractor layer has been frozen, in contrast with the transformer encoder part which has been fine-tuned for the valence-arousal prediction task (as accomplished in [43] and in accordance with [49]). The branching flow diverges into a mean pooling step, with the consequent output to be processed by a 2-layer BLSTM network and a final Dense Regressor for predicting the emotion dimensional values. A second trial development of the nominated model was inspired by two prior works previously mentioned: *Audeering* Wav2vec 2.0 [43] and the CNN-based feature extraction methodology described by Yang and Hirschberg [32]. As noticeable in Figure 2, the feature extraction process is organized into two distinct branches, each influencing complementary characteristics of the audio signal for a comprehensive emotion recognition system. The process begins with the mean and standard normalization by the *Wav2vec Processor* of the input speech waveform, followed by padding to ensure uniform length of 6 seconds. The normalized waveform is then sent through two parallel pathways for feature extraction:

- in the left side, the signal passes directly through the pre-trained Wav2vec2.0-base net, which contains approx-

imately 85M parameters. For the fine-tuning process, the CNN extractor layer has been frozen, whereas the transformer encoder part has been trained for conforming to the valence-arousal prediction task (as accomplished in [43] and in accordance with [49]). The pathway undergoes to a mean pooling step for reducing the high number dimension of the embeddings.

- In the right arm of the architecture, a Mel Fourier Transform and a successive Log-Power conversion is applied to the speech waveform, resulting in a log-power mel spectrogram. The spectrogram is then augmented with first and second order derivatives, transforming into 40-dimensional Mel Scale with 3 channels matrix. The contribution of this stage adds a deep comprehension on small temporal changes in frequency, which are essentials for SER. Subsequently, a 2D CNN is employed for a better insight onto spectral and temporal domains. Specifically, two layers with ReLU activations and max pooling are utilized, with a final flattening.

The two branches are thereafter combined into an unique feature representation, which is consecutively fed into a 2-layer Bidirectional LSTM (BLSTM), as performed in [32]. This layer performs a more coherent and long time dependency with a growth in understanding emotional tendencies. The final output is forwarded to a regressor layer, mapping the learned features to valence and arousal scores. The custom models' training gait and the test set's metrics are illustrated and described in section II-D

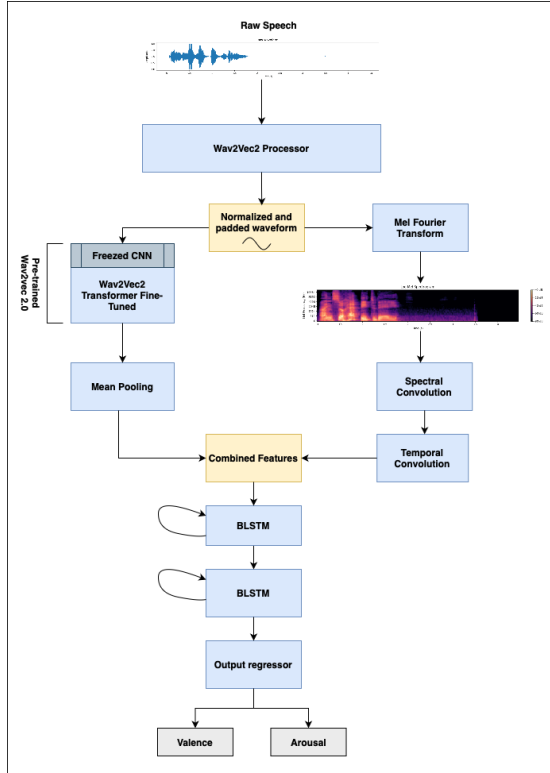


Fig. 2. Custom model architecture (second implementation).

C. Datasets

In this project we find a distinction, apart for valence and arousal attributes, regarding the datasets and their usage in the following work: SER for the fine-tuning phase and song recommendation for the final user. Specifically, the firsts include WAV speech files sampled at 16 Khz, whereas the seconds gather MP3 songs originally sampled at 44.1 Khz.

- SER datasets:

- IEMOCAP [37]: it incorporates approximately 12 hours of audiovisual data involving 10 actors. In addition to plenty of diverse data, its release provides categorical and dimensional annotations for each utterance;
- MuSE-CAR [50]: multi-modal dataset for sentiment analysis with in-the-wild data collection, particularly focused on emotional involvement on automobile advertisements. Moreover, such set is crucial for learning context shades in diverse environments.
- MSP-Podcast [36]: the corpus has 151,654 speaking turns, which makes this dataset the largest naturalistic speech emotional one in the community. Due to its sensible dimensionality, a sample of 30% has been utilized.

- Songs datasets:

- Database for Emotional Analysis in Music (DEAM) [46]: it provides 1802 songs or excerpts with valence and arousal dynamic annotations for a Music Emotion Recognition (MER) goal;
- PMemo [47]: it contains 794 songs comprising emotion annotations;
- SpotiGeM [48]: database which embraces a subset of artists and tracks metadata from Spotify platform.

1) *Data Management*: The subsequent phase preserves the essential information for training and user listening. Mainly, the key features are speech or song files (maintaining only some metadata such as track and artist name), valence and arousal, both normalized between 0 and 1. For SER datasets, the tables of dimensional values and speech files, converted in *Numpy* arrays, are joined by the id given by the dataset. A similar procedure is accomplished for music tracks, but their file-path have been saved instead of converting files in arrays. SpotiGeM actually maintains a preview link for music listening: especially, only when the playlist is generated, songs are fetched by HTTP requests and then conserved in temporary files, avoiding a long-time saving process.

D. Training

The training procedure is customized for optimizing valence and arousal specific prediction with the avail of the Concordance Correlation Coefficient (CCC) loss. The setup starts with splitting the dataset (see II-C) into train (80%) and validation (20%) sets. The considered files are padded with a maximum length of 6 seconds, each processed via Wav2Vec 2.0 Feature Extractor. This filter is applied due to a lack of sufficient GPU memory to process the entire batch

dimensionality for longer contents. However, those seconds are deemed sufficient, since emotions are captured in short time windows as a consequence of their inherent dynamism [51]. Other than this circumstance, we have to take in account the fact that the dimensional values (valence and arousal) are calculated considering an overall average across the speech fragment in almost every record. Thus, contemplating long files would be a risk in terms of accuracy and stability of the model. Furthermore, vocal requests often don't last more than that amount of time, meaning that the duration of the speech extent chosen is valid. Besides, few files are randomly augmented, with the possibility that they could be pitch-shifted or superimposed by either other speech files or background noises. The technique is applied in favor of incrementing contextual simulations across speech situations. The input data is dissected into a batch size of 32, leading to construct each batch with 96000 number of samples and a mel spectral window of $Mel\ features = 3$, $number\ of\ Mels = 48$, $n_FFT = 512$, $window_size = 25\ ms$. The optimization is conducted by ADAMW Optimizer with a starting learning rate of $1e-5$ and adjusted by OneCycleLR scheduler. The loss calculation is conducted by averaging the respective valence and arousal CCC losses for each step and then back-propagated. The train is run for 50 epochs. The model selected is the one that performs best on the validation set with Early Stopping technique after 15 epoch of no improvement. The CCC-loss trend over epochs for the first custom model implementation, as extended in section II-B, is the following:

Fig. 3. Custom model's training plot with no addition of Mel spectrograms in input.



TABLE II
CUSTOM MODEL'S METRICS RESULTS (FIRST IMPLEMENTATION)

Metrics Results (Custom Model: first implementation)		
Metric	Valence	Arousal
CCC Loss	0.715	0.501
L1	0.184	0.140
L2	0.054	0.032
R ²	-0.734	-0.156

As the plot II-D shows, the validation loss decreases to a minimum value around 0.6 after 4 epochs. However, the

generalization in the test set is not brilliant. Consequently, the confirmation takes place for the metrics, as noticeable in the table II. Assuming that mel spectrograms add a richer frequency representation and information, it has been checked whether this feature might improve the performance. Nevertheless, despite it reaches a further minimum around 0.55, the convergence in the plateau is more unstable, upholding the poor metrics obtained in the test set (figure II-D and table III).

Fig. 4. Custom model's training plot with Mel spectrograms in input (second implementation).

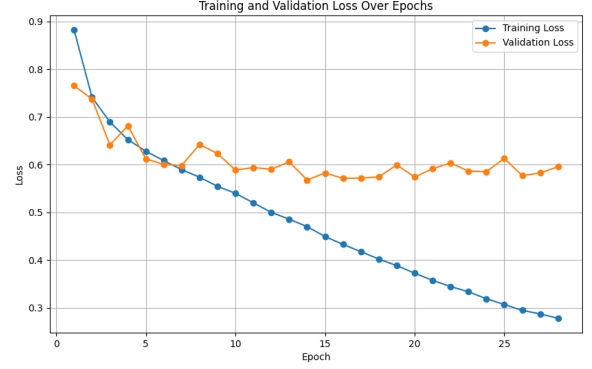


TABLE III
CUSTOM MODEL'S METRICS RESULTS WITH ADDITION OF MEL SPECTROGRAMS IN INPUT.

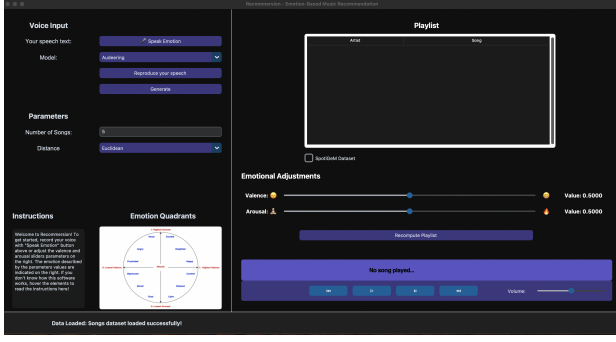
Metrics Results (Custom Model: second implementation)		
Metric	Valence	Arousal
CCC Loss	0.754	0.667
L1	0.145	0.182
L2	0.034	0.048
R ²	-0.075	-0.743

E. Graphic User Interface

As mentioned in the paragraph II-A, Recommerson is a vocal assistant which tries to detect a contextual background relying on emotional user-provided inputs. Its features are seamlessly integrated into a GUI in figure II-E, where users can access instructions by hovering over the interactive widgets and take a cue from the shown emotion quadrant, giving knowledge of how those continuous values shape approximately a certain emotion. The system processes the user's input through a basic Recommender System, by either a microphone ("Speak Emotion" button), which uses a Speech Emotion Recognition (SER) model (*Audeering* or Customized one), or sliders, which are found in the "Emotional Adjustments" GUI section. Then, in the "Parameters" portion, a given similarity (Euclidean or Cosine) is chosen together with a desired number of tracks, resulting in a generated playlist sorted by the emotional input criterion. Any track can be selected in the list located in the left-upper corner. Once the recommendation is produced and a musical fragment begins playing, users gain access to the song's reproduction interface

(in the right below corner). The input dictated by the user can

Fig. 5. GUI of Recommersion



be easily listened anytime by pressing the "Reproduce your speech" button. If the suggestions doesn't reflect the expectations of the user, a recomputation can be easily actuated by simply repeating the process. An user might wonder, for example, whether a particular song feels too happy: in that case, he/she can decrement the valence parameter with the purpose of increasing sadness level. Finally, after making their desired changes, the "Recompute Playlist" button allows users to submit updated emotional parameters, prompting the system to recalculate and refine the recommendations accordingly.

III. EXPERIMENTS

This section explains how the tests were conducted. A group of 19 people, subdivided in 12 males and 7 females with an age from 22 to 63 years old, took part on this analysis responding on a survey. The analysis examines two models, especially *Audeering* and Custom Model (with Mel spectrograms). In particular, the questions aim to comprehend whether the discrepancy between the performance of the two models significantly impacts the recommendation. Furthermore, to investigate which distance criteria is better for this specific purpose. Hence, the experiment was subdivided in:

- 1) experiencing the software broadly;
- 2) recording one specific emotion (fixed for the entire experiment), chosen indifferently by the tester, and place 5 as number of songs;
- 3) generate the music recommendation with *Audeering* model and Euclidean Distance;
- 4) generate the music recommendation with *Audeering* model and Cosine Distance;
- 5) generate the music recommendation with Custom model and Euclidean Distance;
- 6) generate the music recommendation with Custom model and Cosine Distance.

After every step, the users must have listened to every song and answered these questions:

- for each piece, from 0 to 3 (0: non relevant, 1: neutral, 2-3: highly relevant), express the relevance in correspondence with the emotion they entered;
- alter the valence and arousal sliders until it has been reached the user's ideal recommendation and register the difference.

This assay does not include the huge multitude of songs that popular streaming platforms own and the memorization of any listener utilization. Thus, the program will not be able to suggest songs that reflect the taste of the user. Therefore, the experiment has only been based on emotional characters, without taking into account any specific vocal commands. Trial users were completely aware of these peculiarities before starting to examine.

These pointed queries target to obtain specific ranking quality measures of the Recommender System, among others:

- Mean Average Precision (MAP): it evaluates how well a recommender system ranks relevant items higher in the list. Elements ranked 2 or 3 are considered in the ground truth. The calculation is:

$$AP = \frac{1}{R} \sum_{k=1}^N P(k) \cdot \text{rel}(k)$$

where:

- R is the total number of relevant items (ground truth);
- $P(k)$ is the precision at rank k (in our case 5);
- $\text{rel}(k)$ is 1 if the item at rank k is relevant, otherwise 0.

Then MAP is the average of AP across all users:

$$MAP = \frac{1}{U} \sum_{i=1}^U AP_i$$

where U is the total number of users.

- Mean Reciprocal Rank (MRR): it evaluates the position of the first relevant item in the recommendation list (useful when the first relevant suggestion is the most important). The calculation is:

$$MRR = \frac{1}{U} \sum_{i=1}^U \frac{1}{\text{rank}_i}$$

where rank_i is the rank of the first relevant item for user i .

- Normalized Discounted Cumulative Gain (NDCG): it evaluates ranking quality by considering both relevance and position in the ranked list (higher importance to top-ranked relevant items). The calculation is:

$$DCG@K = \sum_{i=1}^K \frac{2^{\text{rel}_i} - 1}{\log_2(i + 1)}$$

where rel_i is the graded relevance (0–3) of the item at position i . The Ideal DCG (IDCG) at rank K is:

$$IDCG@K = \sum_{i=1}^K \frac{2^{\text{rel}_i^*} - 1}{\log_2(i + 1)}$$

where rel_i^* is the ideal relevance of the item at position i (perfectly ranked). Finally, the NDCG at rank K is:

$$NDCG@K = \frac{DCG@K}{IDCG@K}$$

In our case, the IDCG is retrieved by ordering in a decrescent way the ranking list and then averaged by the number of users.

The trial described above was repeated twice, with the only difference in amplifying the number of music fragments using *SpotiGeM* dataset. The distinction lays on the "Arousal" feature, which in this set of data is replaced by the "Energy" column. Therefore, with a distinct measurement for this usage, we can consider the interrelationship of arousal and energy in this singular juncture.

A. Results

In the following outline, an in-depth analysis of the results was conducted. Two main models were considered: *Audeering* and *Custom*, each assessed with both *Euclidean* and *Cosine* distances. The performance of these models was evaluated using the ranking metrics previously mentioned in III: Mean Average Precision (MAP), Mean Reciprocal Rank (MRR) and Normalized Discounted Cumulative Gain (NDCG). Additionally, the incorporation of *SpotiGeM* dataset gives a further analysis of the system's effectiveness.

TABLE IV

THIS TABLE SHOWS, FOR EACH COMBINATION, THE METRICS RESULTS.

Recommender System Metrics Results			
Model / Distance	MAP	MRR	NDCG
Audeering / Euclidean	0.728	0.759	0.805
Audeering / Cosine	0.556	0.619	0.780
Custom / Euclidean	0.661	0.675	0.778
Custom / Cosine	0.547	0.559	0.710

The first investigation is portrayed by Table IV. The highest MAP score of **0.728** was achieved using the *Audeering* model with *Euclidean* distance. Similarly, the best MRR (**0.759**) and NDCG (**0.805**) scores were observed for the same configuration. This confirms the hypothesis in which *Audeering* model performs best under the Euclidean distance. Conversely, a lower performance by the *Custom* model has emerged, with its highest MAP score reaching **0.661** under the Euclidean criterion. The lowest performance was observed for the *Custom* model using *Cosine* distance, with MAP, MRR and NDCG scores of **0.547**, **0.559**, and **0.710**, respectively.

Despite a significant variance regarding SER predictions, these findings suggest that *Audeering* and *Custom* ranking metrics are not too divergent from each other.

Below, Table V illustrates the updated results after integrating *SpotiGeM* dataset, remembering that Arousal and Energy column are equally intended.

This inclusion led to a slight improvement in the *Audeering* model with *Euclidean* distance recommendation, remaining the best-performing configuration. MAP (**0.787**), MRR (**0.816**) and NDCG (**0.842**) scores should not unbalance the assumption whereby emotion energy and song energy account the same. As a matter of fact, *SpotiGeM* amplify around ten times the amount of songs, probably signifying a major granularity correlated to the valence parameter. Remarkably, the performance of the *Custom* model exhibited mixed trends. While its

TABLE V

THIS TABLE SHOWS, FOR EACH COMBINATION, THE METRICS RESULTS WITH THE ADDITION OF THE *SpotiGeM* DATASET

Recommender System Metrics Results (with SpotiGeM)			
Model / Distance	MAP	MRR	NDCG
Audeering / Euclidean	0.787	0.816	0.842
Audeering / Cosine	0.696	0.737	0.743
Custom / Euclidean	0.606	0.585	0.781
Custom / Cosine	0.545	0.566	0.826

MAP and MRR scores decreased slightly under the Euclidean metric (from **0.661** to **0.606** and **0.675** to **0.585**, respectively), the NDCG score remained stable at **0.781**. On the other hand, the *Custom* model with *Cosine* distance demonstrated an improvement in NDCG (from **0.710** to **0.826**). The music fragment expansion confirms the instability of the custom model observed under the test set examination.

B. Discrepancy in Recommendations

The discrepancies between the predictions made by the Speech Emotion Recognition (SER) models and the ideal recommendations obtained using valence/arousal sliders are summarized in Table VI. The values represent the differences between the predicted emotional attributes and the ideal recommendations, meticulously selected by moving the sliders until the playlist satisfies the user almost completely. The calculation is the following:

$$\text{Discrepancy} = \frac{1}{U} \sum_{i=1}^n y_i - \hat{y}_i \quad (1)$$

where:

- U is the total number of users,
- y_i represents the predicted value,
- $\hat{y}_i$ represents the ideal value.

Once again, *Audeering* / *Euclidean* combination showed the lowest discrepancy in valence (**0.00389**) and moderate discrepancy in arousal (**-0.01563**), whereas cosine distance-based models demonstrated higher differences.

TABLE VI

RECOMMENDER SYSTEM DISCREPANCY (BASELINE)

Recommender System Discrepancy		
Model / Distance	Valence Loss	Arousal Loss
Audeering / Euclidean	0.00389	-0.01563
Audeering / Cosine	0.01053	-0.04747
Custom / Euclidean	-0.02842	0.07926
Custom / Cosine	-0.01311	0.00963

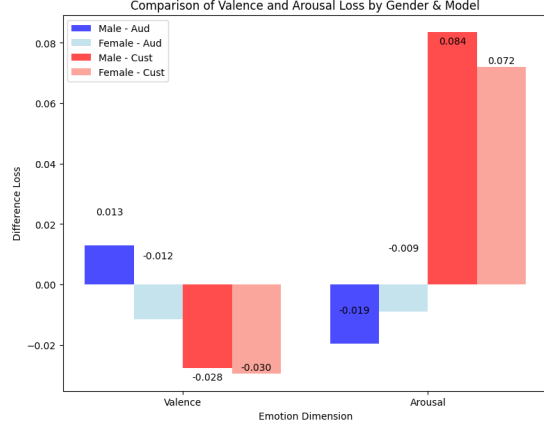
This confirmation takes also place when the number of music fragments is augmented with *SpotiGeM*, as shown in Table VII.

To conclude the analysis, a further comparison on male and female voices has been studied (examining only *Euclidean* similarity criterion). Figure III-B shows that differences between them are almost rejectable in this background, stating that both SER models prognosticate coherently and indistinctly with diverse voice spectrum.

TABLE VII
RECOMMENDER SYSTEM DISCREPANCY (WITH SPOTIGEM)

Recommender System Discrepancy (with SpotiGeM)		
Model / Distance	Valence Loss	Arousal Loss
Audeering / Euclidean	0.01342	-0.01774
Audeering / Cosine	-0.02353	0.02653
Custom / Euclidean	0.01005	0.07905
Custom / Cosine	-0.03511	0.09579

Fig. 6. Histogram comparing gender emotion dimensional discrepancies of Audeering/Custom - Euclidean combination.



IV. CONCLUSION AND FUTURE WORKS

The main power of such CARS, *Recommersion*, is carried out by Speech Emotion Recognition (SER) models and real-time valence-arousal manipulation, supplemented by a straightforward vocal assistant. The crucial challenge, as set forth in Section II-D, was to discover a powerful and robust emotion predictor capable of being so thorough and *Audeering's* contender, in order to affine song suggestions to the maximum extent possible. Nonetheless, the experimental phase exhibited that *Audeering* and *Custom* ranking metrics not deviate by far, although a clear dissimilar testing performing. Indeed, ideal recommendations don't move away consistently in relation to SER ones in both experiments. Another accountable detail is that emotion dimensional labels in songs might suffer flimsy deviations, derived by inconsistent measurements caused by the diverse datasets nature. Thus, despite not contemplating many different ranking cuts and a large number of songs and testers, the use of vocal assistants in this circumstance seems promising. In relation to this, emphasize new trials with performative SER systems and clever recommendation tactics may increase the usability of music platforms, as stated by more than the majority of the testers. Furthermore, in test phase people often imagine car-driving situations: the fusion of frequent music listening and safety awareness under this restrict environment elevate the system in question to be utilized with no concerns, contouring what [22] expressed in her assay. Likewise, this software affirms a new CARS developing possibility under vehicular situations [21]. Effectively, trial users found the project, in a scale from useless to extremely useful, averagely very valuable

as figured in Histogram 7.

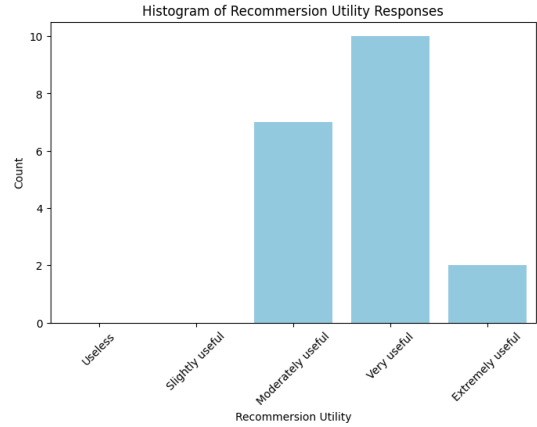


Fig. 7. The figure plots the software utility demanded and answered by users during experiment phasing.

Henceforth, some future avenues can improve the system's usability and accuracy. Considering a cross-correlation of speech and text emotion can target a further specific context. An hypothetical situation could point to a request from the user which demands cheering music despite his sad mood. In addition, although plenty of users could not understand foreign words, the incorporation of lyrics sentiment analysis with SER can give an overwhelming and conveyed music emotional experience. For allowing such feature, operating in large-scale music platforms with open APIs can clearly facilitate songs recommendation. As a matter of fact, negligible differences in valence or arousal values can outline a stand-alone feeling. Thus, contemplating a vast tune catalog can lead to a completely whole, different and accurate suggestion. In reference to this preciseness, sliders assist in the meticulous research of the ideal playlist, especially when the prediction of SER models is inaccurate. With such purpose, songs finding can be enhanced by integrating an emotion categorical selector. This filter can shrink the range in which specific values lay on that exact emotion, easing the consumers in their lookup. Furthermore, practically every testers decreed that rhythm is the determinant factor for judging songs in an emotion sphere. Its key role may adjust the training network architecture: contemplating rhythm-based features such as tempo, beat histogram and syncopation patterns could have improved prediction accuracies and suggest more refined music fragments. Dealing with a wide assortment head to an User-Value (UV) Decomposition [52] for incorporating user preferences and historical data. This matrix factorization technique help to learn latent preferences providing personalized suggestions. Such aspect could have facilitate the experiment phase, in particular with an increment of the ranking metrics quality. Alternatively, diversity, serendipity and novelty metrics would have taken into consideration. The balance among them could introduce listeners to new and unexpected music while still aligning with their tastes.

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